

On the Convergence of Stochastic Gradient Descent under Heavy-Tailed Noise

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Abstract

We revisit the convergence theory of stochastic gradient descent (SGD) in the regime where the gradient noise has only finite p -th moment for some $p \in (1, 2)$. In this heavy-tailed regime the classical $O(1/\sqrt{n})$ rate for convex objectives no longer holds, and the step-size schedule that achieves the best attainable rate departs from the Robbins-Monro recipe of $\eta_k = \Theta(1/k)$. We prove two main results: a tight upper bound of $O(n^{-(p-1)/p})$ on the optimisation gap of clipped SGD with a polynomial step-size schedule, and a matching lower bound under a fairly mild adversarial-noise oracle. Numerical experiments on a synthetic generalised-linear model with Student-t gradient noise confirm the rates to within small constants and illustrate the catastrophic failure mode of vanilla SGD without gradient clipping. We close with practical recommendations for adaptive-clipping schemes and a discussion of the implications for high-dimensional optimisation on noisy hardware.

1. Introduction

Stochastic gradient descent is the unrivalled workhorse of modern large-scale optimisation, underpinning everything from logistic regression on tabular data to the pre-training of trillion-parameter language models. Its convergence theory under *light-tailed* noise — that is, when the stochastic gradient has a finite variance and the oracle returns samples with sub-Gaussian or at least sub-exponential concentration — is now classical and rests on a small handful of inequalities tracing back to Robbins and Monro [1], Polyak [2], and Nemirovsky and Yudin [3]. For a convex Lipschitz objective on a bounded domain, the iterates of vanilla SGD with step size $\eta_k = \Theta(1/\sqrt{k})$ achieve an optimisation gap of $O(1/\sqrt{n})$ after n iterations, and this rate is known to be unimprovable in the worst case [4].

The picture changes dramatically when the gradient noise has only a finite p -th moment for some $p \in (1, 2)$ — a regime that is far from theoretical: gradient distributions in deep-learning workloads have been documented as heavy-tailed [Simsekli2019; Gurbuzbalaban2021], and the recent rise of low-precision and stochastic-rounding hardware introduces noise that is bounded but skewed and only weakly concentrated. In this *heavy-tailed* regime several previously innocuous algorithmic choices become load-bearing. Step-size schedules designed for the light-tailed regime can drive the iterates arbitrarily far from the optimum after a single bad sample; gradient clipping, which is a *practical* fix that virtually every deep-learning practitioner already deploys, turns out to be a *theoretical* necessity as well.¹

combat the exploding-gradient problem in recurrent network training [5] long before its role in heavy-tailed convergence was understood. Its theoretical justification under heavy-tailed noise was clarified by Zhang et al. [6] and Gorbunov et al. [7], whose analyses we extend in this paper.

¹Gradient clipping was introduced as a heuristic to

The questions we take up here are: *what is the optimal achievable rate for SGD under finite- p -th-moment gradient noise, and what step-size and clipping schedule attain it?* Our two main results, stated informally:

1. **Upper bound (Theorem @thm:upper).** Clipped SGD with a polynomial step-size schedule $\eta_k = \eta_0 k^{-1/p}$ and clipping threshold $\tau_k = \tau_0 k^{1/p}$ achieves an optimisation gap of $O(n^{-(p-1)/p})$ on convex Lipschitz objectives with p -th-moment noise.
2. **Lower bound (Theorem @thm:lower).** No first-order stochastic algorithm, clipped or not, can improve on $\Omega(n^{-(p-1)/p})$ over the class of heavy-tailed instances we consider.

Together the two results pin the minimax rate at $\Theta(n^{-(p-1)/p})$ and identify clipped SGD with a polynomial schedule as a minimax-optimal algorithm. At $p = 2$ we recover the familiar $O(1/\sqrt{n})$ rate; at $p = 1.5$ the rate is $O(n^{-1/3})$, substantially slower; and as $p \downarrow 1$ the rate degenerates to a logarithm of n , which is the best one can hope for under barely-finite mean noise.

The remainder of the paper is organised as follows. Section 2 reviews the relevant background on heavy-tailed concentration and fixes notation. Section 3 develops our method — the clipped SGD algorithm and its associated step-size schedule. Section 4 proves the upper and lower bounds and reports numerical experiments on a synthetic generalised-linear model. Section 5 discusses implications for practice, with a focus on adaptive clipping. Section 6 concludes. Section 7 collects references and definitions.

2. Background

2.1. Heavy-tailed concentration

Let X be a real-valued random variable with $\mathbb{E}|X|^p < \infty$ for some $p \in (1, 2]$, and let $X_1, \dots, X_n$ be independent copies of X . The classical central-limit theorem guarantees that $n^{-1/2} \sum_k (X_k - \mathbb{E} X) \rightarrow \mathcal{N}(0, \sigma^2)$ in distribution when $p = 2$, but no such universal limit exists for $p < 2$: the appropriate limit is an α -stable law with index $\alpha = p$, and the rate of convergence is governed by the tail behaviour of X .¹

function $\phi(t) = \exp(-|t|^\alpha)$ for some scale $c > 0$ and satisfies $\mathbb{E}|Y|^q < \infty$ iff $q < \alpha$. When $\alpha = 2$ it is the Gaussian; when $\alpha = 1$ it is the Cauchy. See Nolan [8] for a modern treatment.

For finite-sample bounds the relevant tool is the *truncated moment inequality*: for any threshold $\tau > 0$ and any $p \in (1, 2]$,

$$\Pr[|S_n - \mu| \geq \tau] \leq \frac{C \mathbb{E}|X|^p}{n^{p-1} \tau^p},$$

where $S_n = \sum_k X_k$, $\mu = \mathbb{E} X$, and C is a universal constant. This is the heavy-tailed analogue of Hoeffding's inequality, and it is the inequality we will deploy repeatedly in the proof of Theorem @thm:upper.

¹An α -stable distribution has characteristic

2.2. Stochastic optimisation under heavy-tailed noise

The earliest treatments of heavy-tailed stochastic optimisation are due to Nemirovsky and Yudin [3] in the offline-oracle setting and to Polyak and Tsykin [9] in the online setting; both works essentially assume finite variance and adapt the classical proofs. The recent literature on truly heavy-tailed noise begins with Zhang, Karimireddy, et al. [6], who proved an $O(n^{-(p-1)/(3p-2)})$ rate for clipped SGD on smooth strongly convex objectives; this rate is suboptimal but holds without an exact moment constant. Gorbunov et al. [7] improved this for general convex objectives to $O(n^{-(p-1)/(2p-1)})$, and Cutkosky and Mehta [10] obtained the rate $O(n^{-(p-1)/p})$ that we match in this paper, via a different (and more delicate) analysis based on the implicit gradient method. Our contribution is a direct analysis of explicit clipped SGD with a polynomial step-size schedule, which is the algorithm that practitioners actually run.

3. Method

3.1. Notation and assumptions

Let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be a convex objective and let $x^* \in \arg\min f$ be a minimiser. We assume f is G -Lipschitz on a convex bounded domain $\mathcal{X}$ of diameter D , and that we have access to a *stochastic first-order oracle* $\mathcal{O}$ which, when queried at a point $x \in \mathcal{X}$, returns a random vector $g \in \mathbb{R}^d$ satisfying

$$\mathbb{E}[g \mid x] = \nabla f(x), \quad \mathbb{E}[\|g - \nabla f(x)\|^p \mid x] \leq \sigma^p,$$

for some $p \in (1, 2]$ and some $\sigma > 0$. No higher-order moments are assumed. When $p = 2$ this is the standard bounded-variance assumption; when $p < 2$ we are in the heavy-tailed regime.

3.2. Clipped SGD with polynomial schedule

The algorithm we analyse is the following clipped-SGD scheme. Fix $x_0 \in \mathcal{X}$, step-size sequence $\eta_k > 0$, and clipping threshold sequence $\tau_k > 0$. For $k = 0, 1, 2, \dots$:

1. Query the oracle at x_k to obtain g_k .
2. Form the clipped gradient $\tilde{g}_k = \text{clip}_{\tau_k}(g_k)$, where $\text{clip}_{\tau}(v) = v \cdot \min\{1, \tau / \|v\|\}$.
3. Update $x_{k+1} = \Pi_{\mathcal{X}}(x_k - \eta_k \tilde{g}_k)$, where $\Pi_{\mathcal{X}}$ is the Euclidean projection onto $\mathcal{X}$.

The output is the *uniformly averaged iterate* $\bar{x}_n = \frac{1}{n} \sum_{k=0}^{n-1} x_k$.

The schedules that achieve the optimal rate are $\eta_k = \eta_0 \cdot (k+1)^{-1/p}$ and $\tau_k = \tau_0 \cdot (k+1)^{1/p}$, with the constants η_0 and τ_0 to be chosen as functions of D, G, σ , and p alone. Both schedules deviate from the Robbins-Monro recipe of $\eta_k = \Theta(1/k)$ that is optimal under light-tailed noise — as we will see in the proof, the polynomial schedule is precisely what is needed to balance the bias introduced by clipping against the variance of the unclipped tail.

4. Results

4.1. Upper bound

Our first main result is the following.

Theorem

1 (Upper bound).

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex, G -Lipschitz on a convex bounded domain $\mathcal{X}$ of diameter D , and suppose the stochastic first-order oracle satisfies $\mathbb{E} [\|g - \text{grad } f(x)\| \mid \mathcal{H}_k] \leq \sigma$ for some $\sigma \in (0, 1]$. Then clipped SGD with $\eta_k = \eta_0 / (k+1)$ and $\tau_k = \tau_0 / (k+1)$ satisfies

$$\mathbb{E} [f(\bar{x}_n) - f(x^*)] \leq C \cdot n^{-(q-1)/q}$$

where $C = C(D, G, \sigma, q)$ is a constant depending only on the listed parameters, with η_0 and τ_0 chosen as polynomial functions of those parameters.

[#thm:upper]

Proof sketch. Let r_k denote $x_k - x^*$. Expanding the recursion $x_{k+1} = \Pi(x_k - \eta_k \tilde{g}_k)$ and using non-expansiveness of the projection Π , the standard descent inequality bounds $\|r_{k+1}\|^2$ above by $\|r_k\|^2 - 2\eta_k \langle \tilde{g}_k, r_k \rangle$ plus the noise term $\eta_k^2 \|\tilde{g}_k\|^2$, for every $k \geq 0$.

Taking conditional expectations, the cross term decomposes into an exact-gradient piece and a *clipping bias* piece. The exact-gradient piece equals $\langle \nabla f(x_k), r_k \rangle$, which the convexity of f bounds below by $f(x_k) - f(x^*)$. The clipping bias is controlled by the truncated moment inequality of Section 2.1 and is $O(\sigma^p / \tau_k^{p-1})$. The final noise term $\eta_k^2 \mathbb{E} \|\tilde{g}_k\|^2$ is bounded by $\eta_k^2 (G^2 + \tau_k^2)$ via a separate truncation argument.

Summing from $k=0$ to $n-1$, dividing by $\sum \eta_k$, and applying Jensen's inequality to the averaged iterate $\bar{x}_n$, the telescoping leaves three error terms whose orders in n are $D^2 / \sum \eta_k$, $\sigma^p \sum \eta_k / \tau_k^{p-1}$, and $\sum \eta_k^2 (G^2 + \tau_k^2)$. Plugging in $\eta_k \propto k^{-1/p}$ and $\tau_k \propto k^{1/p}$, each of the three sums evaluates to $\Theta(n^{-(p-1)/p})$, and dividing by the harmonic-like sum of step sizes (which is $\Theta(n^{-(p-1)/p})$ as well) yields the claimed rate. The full computation, including the choice of the constants η_0 and τ_0 that balance the three terms, is in Appendix A of the technical report [11]. $\square$

A few remarks on the proof. First, the rate $n^{-(p-1)/p}$ is attained simultaneously by all three error sources; this is no coincidence but the consequence of the deliberate balancing of η_k and τ_k . Second, the schedule is *non-adaptive*: it depends on p only through the exponents. An adaptive variant that estimates p on the fly is sketched in Section 5.

4.2. Lower bound

The matching lower bound is the following.

Theorem

2 (Lower bound).

For every $q \in (1, 2]$ and every $n \geq 1$, there exists a convex Lipschitz function f on a bounded domain and a heavy-tailed stochastic first-order oracle satisfying the q -th-moment assumption of Section 3, such that any first-order stochastic algorithm, clipped or otherwise, has an expected optimisation gap of at least

$$c(q) \cdot n^{\sup\{-(q-1)/q, -1\}}$$

after n oracle queries, where $c(q) > 0$ is a universal constant depending only on q .

[#thm:lower]

Proof sketch. We use a hard one-dimensional instance: $f(x) = |x|$ on $[-D, D]$, $x^* = 0$. The oracle returns $g = \text{sign}(x)$

- ξ_i , where ξ_i is sampled from a Pareto distribution with

parameter p (so that $\mathbb{E}|\xi_i|^q < \infty$ iff $q < p$). The sign of the gradient is occasionally flipped by a heavy-tailed shock in ξ_i ; any algorithm that follows the (noisy) gradient will overshoot. A Le Cam two-point reduction — testing $f(x) = |x|$ against $f(x) = |x - \epsilon|$ — then gives a lower bound that matches the upper bound up to the constant $c(p)$. Details are in Appendix B of [11]. $\square$

Together, Theorems @thm:upper and @thm:lower pin the minimax rate at $\Theta(n^{-(p-1)/p})$. Note that the lower bound holds even for algorithms that *know* the value of p ; an adaptive algorithm that estimates p from the oracle history can therefore lose at most a constant factor.

4.3. A figure and a table

The empirical convergence rates of clipped SGD on a synthetic generalised-linear model with Student-t gradient noise are summarised in Table 4.1 and illustrated in Figure ?. The problem is logistic regression on $d = 50$ Gaussian features with $n = 10^4$ training samples, and the oracle’s noise was added by contaminating each mini-batch gradient with a Student-t random variable of degrees of freedom $\nu = 1 + 1/(p - 1)$, calibrated so that the resulting noise has finite p -th moment and infinite higher moments.

p	Theory rate	Empirical slope	Const. C
1.25	$n^{-0.20}$	-0.21	2.84
1.50	$n^{-0.33}$	-0.32	1.97
1.75	$n^{-0.43}$	-0.42	1.51
2.00	$n^{-0.50}$	-0.49	1.18

Table 4.1.

The empirical slopes match the theoretical exponents to within two percent across the four values of p , well within the noise of a single training run averaged over 32 seeds. The constant C shrinks monotonically with p , consistent with the prediction of Theorem @thm:upper that $C(D, G, \sigma, p) \propto G \cdot D$ as $p \rightarrow 2$.

[SVG file trapezoid_convergence.svg omitted in non-SVG back-end]

Figure ?.

Log-log plot of the optimisation gap $f(\bar{x}_n) - f(x^*)$ against iteration count n for clipped SGD with the polynomial schedule of Theorem @thm:upper, on the synthetic logistic-regression problem with Student-t gradient noise. Each curve is averaged over 32 seeds; shaded bands show one-sigma intervals. Slopes match Table 4.1 to two percent across the four values of p .

4.4. A diagnostic experiment

To verify that the polynomial schedule of Theorem @thm:upper is not merely *sufficient* but actually *necessary* under heavy-tailed noise, we re-ran the $p = 1.5$ experiment with three competing schedules:

1. **Vanilla SGD with $\eta_k = 1/\sqrt{k}$, no clipping.** The iterates diverged: after $n = 10^4$ iterations the empirical optimisation gap was 1.4×10^2 , four orders of magnitude worse than the clipped baseline.
2. **Robbins-Monro schedule $\eta_k = 1/k$, with clipping.** The iterates converged but at a rate of approximately $n^{-0.17}$, noticeably slower than the theoretical $n^{-0.33}$.
3. **Polynomial schedule of Theorem @thm:upper.** Recovered the theoretical rate to within two percent, as reported in Table 4.1.

The first experiment confirms that *clipping is necessary*; the second confirms that the *polynomial schedule* is also necessary, not merely a convenient choice. Together they validate the algorithmic prescription of Theorem @thm:upper as the right object to implement in practice.

5. Discussion

The bounds derived in Section 4 sharpen three observations and expose one less-discussed subtlety.

First, **gradient clipping is theoretically necessary, not just a heuristic.** Vanilla SGD without clipping has an unbounded expected optimisation gap whenever the gradient noise has infinite variance, because a single bad sample can push the iterate arbitrarily far from the optimum. This is not a matter of constants: the failure is catastrophic and qualitative. The clipping threshold τ_k acts as a *trust region* on the per-step update, and its polynomial growth ensures that the trust region eventually exceeds the noise scale.

Second, **the step-size schedule must be polynomial.** The Robbins-Monro recipe of $\eta_k = \Theta(1/k)$ that is optimal under light-tailed noise is too aggressive under heavy-tailed noise: it shrinks the step size faster than the clipping bias can be tolerated, and the resulting rate is worse by a factor of $n^{(2-p)/p}$.

Third, **the minimax rate $n^{-(p-1)/p}$ is fully described by the moment exponent p .** Specifically, the dependence on the moment constant σ^p enters only through the constant $C(D, G, \sigma, p)$ and not through the exponent. This is the heavy-tailed analogue of the light-tailed result that the variance enters the $O(\sigma/\sqrt{n})$ rate only as a constant.

The subtler observation, mentioned in passing in Section 4.3, is that **adaptive estimation of p is possible but tricky.** A natural adaptive scheme is to estimate $\widehat{p}_k$ from the empirical distribution of $\|g_k\|$ via a Hill estimator [12].[3]¹ The challenge is that the Hill estimator itself has a slow convergence rate of

$O(k^{-1/2})$), which translates into a polylogarithmic loss in the final rate. Whether the polylogarithmic loss is fundamental or an artefact of the Hill estimator remains an open question.

distributions whose tail is exactly a regularly-varying function of finite order, and the bias can dominate the variance for moderate sample sizes. A bias-corrected variant due to Drees [13] mitigates this but introduces an additional tuning parameter.

5.1. Implications for software libraries

Three concrete recommendations for practitioners follow.

1. **Always clip the per-sample gradient, not the per-batch gradient.** The minimax rate of Theorem @thm:upper depends on clipping the *individual* stochastic gradient, not on clipping the mean of a mini-batch. This is because the heavy-tailed tail of a single sample is what drives the failure mode, and averaging within a mini-batch does not change the tail exponent.
2. **Tune the clipping threshold polynomially in k , not as a fixed constant.** Most deep-learning libraries implement clipping with a fixed threshold, which is theoretically suboptimal: a fixed threshold combined with a vanishing step size produces a sub-Robbins-Monro rate of $O(n^{-(p-1)/(p+1)})$ rather than the minimax-optimal $O(n^{-(p-1)/p})$.
3. **Track empirical effective sample size, not the loss alone.** The loss trajectory of clipped SGD under heavy-tailed noise is characteristically jagged — much more so than under light-tailed noise. A robust convergence diagnostic should track a moving median or trimmed mean of the loss rather than the raw running average.

6. Conclusion

We have characterised the minimax-optimal convergence rate of stochastic gradient descent under finite- p -th-moment gradient noise for $p \in (1, 2]$, and we have shown that clipped SGD with a polynomial step-size and clipping schedule attains this rate. The results unify and tighten a sequence of recent advances [zhang2020; gorbunov2020; cutkosky2021] and provide explicit constants suitable for practical implementation. A natural follow-up is the *strongly convex* and *non-convex* settings, where the combination of curvature with heavy-tailed noise raises subtle questions about the trade-off between bias and variance that we have not addressed here. We leave these to future work.

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